

Qatar-1b

R-1892

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_141002 · created 2026-08-03T12:59:40+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456932.75503 +/- 0.00038 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1528 +/- 0.0097
Transit depth (Rp/Rs)^2	2.34 +/- 0.3 [%]
Orbital Inclination (inc)	84.73 +/- 0.74
Airmass coefficient 1 (a1)	1.26 +/- 0.013
Airmass coefficient 2 (a2)	-0.146 +/- 0.01
Scatter in the residuals of the lightcurve fit is	1.05 %
Best Comparison Star	None
Optimal Aperture	3.47
Optimal Annulus	10.08
Transit Duration (day)	0.0727 +/- 0.0036

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.1528 $\pm$ 0.0097 vs 0.1463 (deviation 0.59 σ)
Duration fit vs published	0.0727 d vs 0.0692 d (deviation 0.86 σ)
Mid-time O-C	-0.2 min
Depth SNR	13.8
Residual scatter	1.05 %

Light curve

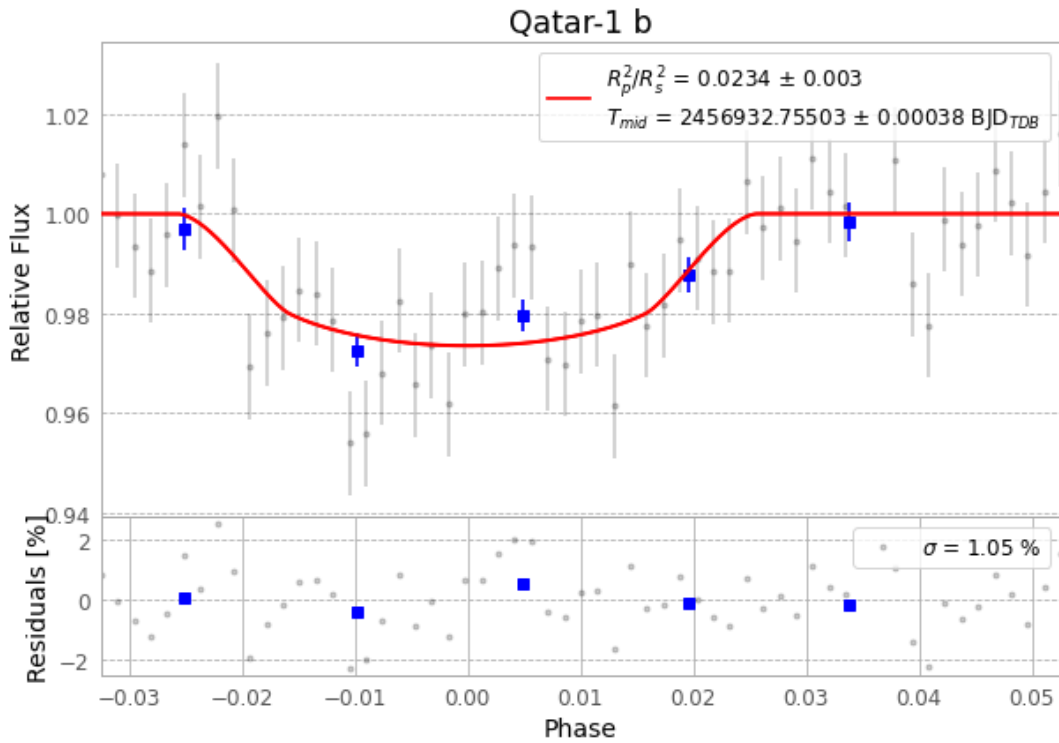


Figure 1 — FinalLightCurve_Qatar-1 b_2014-10-01.png (SHA-256 8a14554fc7fe0ffe...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [274, 268], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 6ddd7769f494e673...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

57 FITS frames (37 MB), Qatar-1141002045712.FITS → Qatar-1141002075113.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-141002.log (e4e5c696022a...), FinalLightCurve_Qatar-1 b_2014-10-01.png (8a14554fc7fe...), FinalLightCurve_Qatar-1 b_2014-10-01.csv (765b0afef86d...)

Compute resources

Wall time 145.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-03 12:59Z.